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# Optimization of a digital health intervention to enhance well-being among adolescent and young adult cancer survivors: Design and methods of the EMPOWER full factorial trial

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## Abstract

Adolescent and young adult cancer survivors (AYAs) experience clinically significant distress and have limited access to supportive care services. Interventions to enhance psychological well-being have improved positive affect and reduced depression in clinical and healthy populations and have not been routinely tested in AYA survivors. We are optimizing a web-based positive skills intervention for AYA cancer survivors called Enhancing Management of Psychological Outcomes With Emotion Regulation (EMPOWER) by: (1) determining which intervention components have the strongest effects on well-being and (2) identifying demographic and individual difference variables that mediate and moderate EMPOWER's efficacy. EMPOWER is a five-session online intervention that teaches behavioral and cognitive skills for increasing psychological well-being. Guided by the Multiphase Optimization Strategy (MOST), we assign two levels (yes, no) to each of five intervention components (positive events, capitalizing, & gratitude; mindfulness; positive reappraisal; personal strengths & goal-setting; acts of kindness), allowing us to evaluate the effects of individual and combined intervention components on positive affect in a full factorial design. Post-treatment AYA cancer survivors (N=352) are recruited from participating NCI-designated comprehensive cancer centers and randomized to one of 32 experimental conditions. Our primary outcome is positive affect; potential mediating and moderating variables include coping self-efficacy and emotional support, respectively. Upon trial completion, we will have an optimized, digital health intervention to enhance psychological well-being among AYA cancer survivors. EMPOWER will be scalable and primed for a large, multi-site trial among AYAs who would otherwise not have access to supportive care interventions to manage distress and enhance well-being.

### Keywords

cancer; psychosocial; well-being; adolescent & young adult; factorial design

## 1. Introduction

Adolescent and young adult cancer survivors (AYAs) are a vulnerable and underserved population at risk for significant psychological distress. There are approximately 90,000 new diagnoses of cancer annually among AYAs (ages 15–39) [1], and more than 2.1 million survivors of AYA cancers in the United States [2]. Five-year survival rates of AYAs exceed >85% [3] yet AYAs are at significant risk for many post-treatment comorbidities including a higher prevalence of asthma, chronic obstructive pulmonary disorder, stroke, and diabetes among AYA cancer survivors, [4, 5] and a greater than two-fold increased rate of cardiomyopathy, stroke, premature ovarian failure, chronic liver disease, and renal failure when compared to AYAs without cancer. [6, 7] AYAs have approximately 3 to 6 decades of life expectancy remaining,[8] underscoring the importance of post-treatment medical surveillance and survivorship care. AYAs face unique challenges given the physical, cognitive, and psychosocial developmental milestones disrupted as a result of cancer and its treatment.[9, 10] They are at greater risk of psychological distress compared to age-matched peers without a history of cancer [11–14] and compared to older adults (ages ≥40) with

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cancer.[15–18] Accordingly, AYAs can benefit from targeted, supportive care interventions to decrease distress and enhance well-being as they navigate post-treatment survivorship.

The National Cancer Institute (NCI) has called for supportive care interventions to address psychological concerns among AYAs [19], and a growing number of interventions are being developed and tested to address AYAs' unique and diverse needs (see [20–23] for relevant reviews). Among the interventions to address psychological concerns in AYAs, the primary focus has been on distress reduction, neglecting to attend to or evaluate potential differential effects on other dimensions of psychological well-being or explicitly targeting them as primary endpoints. A broader and more holistic perspective of mental health acknowledges both the "negative" or distressing aspects as well as the "positive" or life-affirming aspects of the human experience.

Theoretically and conceptually, psychological well-being includes positive affect, meaning and purpose, life satisfaction, and general self-efficacy.[24] Psychological well-being is significantly associated with better health outcomes (better physical health [25], lower risk of mortality)[26–30], is unique from distress, and includes domains that are inherently valued by people with cancer [31]. As evidenced by a recent trial, [32–35] designing interventions to target psychological well-being of AYAs has the potential to impact both positive (i.e., well-being) outcomes as well as negative (i.e., distress) outcomes. However, efficacy data for interventions designed to promote psychological well-being still lack trials targeting distress reduction, creating significant gaps in the clinical armamentarium for AYAs.

Notably, intervention studies to reduce psychological distress among AYAs have primarily focused on acceptability, feasibility, and safety [23, 36, 37]. Digital health interventions represent promising options for survivor engagement, especially with "digital natives" such as AYAs. [38, 39] Given the ubiquitous nature of smartphones among AYAs and their comfort level with and preference for remotely delivered interventions [40], the increasing use of digital health modalities with AYAs is a potential strength to enhance high fidelity intervention delivery, increase reach, and optimize intervention content [39].

Our team developed a novel, multicomponent intervention to enhance psychological well-being that shows promise among people with chronic illnesses [41–44]. We piloted the intervention for AYAs in a digital health delivery format (EMPOWER: Enhancing Management of Psychological Outcomes With Emotion Regulation) [45]. EMPOWER is a five-session online intervention that teaches behavioral and cognitive skills for increasing psychological well-being. Each of these sessions, or components, are developed from existing, evidence-based positive psychological interventions and focused on: (1) positive events, capitalizing, & gratitude; (2) mindfulness; (3) positive reappraisal; (4) personal strengths & goal-setting; and (5) acts of kindness. Preliminary data from our single-arm, multi-site trial suggest EMPOWER is feasible and acceptable to AYAs. In addition, results support proof of concept for reducing distress and improving psychological well-being in the immediate and post-intervention, one-month follow-up period [45].

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The purpose of our current project is to optimize EMPOWER for AYAs by determining which components are effective in promoting well-being. This is an important and innovative aspect of our work; while many psychosocial interventions are inherently multi-component and informed by multi-dimensional frameworks (including positive psychological interventions [46, 47]), few are tested in such a way as to allow for the determination and distinction of which components are actually influencing outcomes. Our trial is designed to address this very question and advance the field in a more efficient manner.

In addition, informed by Fredrickson's Broaden and Build Theory and Revised Stress and Coping Theory [48–50], we will explore mechanisms (i.e., mediators) through which EMPOWER may yield benefits for AYAs and identify which subgroups of AYAs (i.e., moderators) may be most "at risk" and thus more likely to benefit from our intervention. This is another significant benefit of our work. Identifying mediators and moderators contributes to clinician ability to be more precise in the application and use of an intervention with survivors; it provides evidence as to which AYAs may be more likely to benefit from the intervention and informing downstream precision supportive care

To address these questions, utilize resources efficiently, and prepare for a future fully powered randomized controlled trial (RCT), we are using an innovative methodological framework, the Multiphase Optimization Strategy (MOST) [51–53]. MOST designs consist of efficient experiments on distinct intervention components to determine which make the most meaningful contribution to the primary outcome. In our trial, these data inform assembly of a multicomponent intervention to enhance positive affect with minimal resource consumption and participant burden. To our knowledge, this trial is the first to optimize an intervention with a primary focus on promoting psychosocial well-being among AYA cancer survivors.

## 2. Materials and Methods

This project is supported by the National Cancer Institute of the National Institutes of Health under award number R01CA242849 and is registered on ClinicalTrials.gov (NCT04317417). The study procedures described are approved by the Wake Forest University School of Medicine Institutional Review Board and, through reliance agreements, the Northwestern University and MD Anderson Institutional Review Boards.

### 2.1. Study Aims

This study has two specific aims: 1) optimize EMPOWER using a MOST framework to identify which of five intervention components contribute meaningfully to psychological well-being among post-treatment, AYA cancer survivors, and 2) evaluate mediators and moderators of intervention component efficacy for the primary well-bring outcomes.

### 2.2. Study Population

Recruitment began in the December 2021 and will continue through December 2024 at Atrium Health Wake Forest Baptist Comprehensive Cancer Center (AHWFBCCC), Robert H. Lurie Comprehensive Cancer Center of Northwestern University (RHLCCC),

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Levine Cancer Institute and Levine Children's Hospital (LCI/LCH), and MD Anderson Comprehensive Cancer Center (MDACCC). Participant inclusion criteria are: 1) ability to read and understand English, 2) past history of a cancer diagnosis (excluding basal cell skin carcinoma), 3) 15 to 39 years of age at diagnosis of first cancer and currently between the ages of 15–39, 4) currently within 0 to 5 years post-active treatment from initial diagnosis or recurrence (maintenance treatment is permissible), 5) internet access through a home or public shared computer, tablet, or smartphone, and 6) able to provide informed consent. Exclusion criteria included 1) previously hospitalized for a psychiatric illness, or 2) currently receiving end-of-life care.

**2.2.1. Survivor Identification—**Cancer survivors are identified as potentially eligible through review of the electronic health record (EHR) or through prior consent to receive information about relevant research study opportunities in AYA survivorship. IRB-approved recruitment messages/flyers are sent via email, patient portals, or included in AYA program e-newsletters. Direct referral from oncology providers is also used. Based on our team's history recruiting post-treatment AYA survivors and informed by our power analyses, we anticipate a 13% recruitment rate to reach our target accrual goal of 352 AYAs. Notably, we are making intentional efforts to prioritize a diverse sample of AYAs by using inclusive images and language in our recruitment materials (informed by an AYA patient advocate and by our Office of Community Outreach and Engagement at the AHWFBCCC) and through direct outreach via email and patient portal to AYAs who have self-identified as either a member of a racial or ethnic minority group.

## 2.3. Study Procedures

Once a potentially eligible survivor is identified in the EHR, a recruitment email or MyChart message is sent to the survivor outlining details of the study and a link to the screening questionnaire, and if eligible, navigated to the online consent form and baseline study questionnaires. Those who are ineligible are shown a message thanking them for their interest and informing them they are not eligible for the study.

Once consented, enrolled participants are given a unique study ID to complete study questionnaires in REDCap [54, 55]. At completion of the baseline questionnaire and guided by MOST, we assigned two levels (yes, no) to each of the five, weekly intervention sessions ($2^5$). This allows us to evaluate individual and combined intervention components in a classic full factorial design with 32 (2×2×2×2×2) experimental conditions (see Table 1). Based on experimental condition, AYAs receive 0, 1, 2, 3, 4, or 5 EMPOWER components and a corresponding number of 5, 4, 3, 2, 1, or 0 inert content components.

A total of 352 participants are randomly assigned to one of the 32 experimental conditions, resulting in 11 participants per condition. Randomization is pre-programmed, automatic, and monitored by the study team. Block randomization stratified by site is used to ensure balanced group sizes by site at the end of the trial. Participants are blinded to their assignment to intervention or inert content for weekly components during intervention delivery. For study administration purposes, staff members are unblinded. However, all

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outcome measures are participant-reported, potentially mitigating the consequences of staff unblinding on the outcome assessment.

### 2.4. EMPOWER Intervention and Inert Content

The EMPOWER intervention is a five-session web-based intervention that teaches behavioral and cognitive skills for increasing psychological well-being using the following components: Week 1) noting daily positive events, capitalizing on or savoring positive events, and gratitude; Week 2) mindfulness; Week 3) positive reappraisal; Week 4) focusing on personal strengths, setting and working toward attainable goals; and Week 5) small acts of kindness. [45] The individual components we identified for EMPOWER are drawn from evidence-based research and positive psychotherapeutic approaches and originally developed as treatments for depression, but also have benefits for other aspects of psychological well-being.[50]

All participants receive weekly material for a total of five weeks. Weekly material consists of one to two days of written or video didactic content and several days of brief, real-life skills practice, and reporting, with each day's "home practice" taking approximately 10–15 minutes to complete. Participants are instructed to begin each week by reading about each skill and then doing a short exercise to practice the skill and learn how to use it in their lives. They are encouraged to come back to the website daily to practice and strengthen their skills (Figure 1). Daily skills practice is asynchronous and recommended but not required. Participants cannot skip ahead, but they can return to lessons or skills practices they have already completed if they choose. All skills practices are in a "diary" format in which participants' past responses are displayed as a list of exercise dates. Clicking on each date takes them to the responses for that date, so they have a growing list of past positive experiences and practices. Participants are permitted up to eight weeks to complete this self-guided content to account for normal interruptions and unexpected delays in life.

For subjects who do not receive EMPOWER components or only receive some EMPOWER components, inert content is programmed into the web-based intervention delivery and includes lessons and "home practice." Inert content was drawn from a list of topics relevant to AYA survivors to reduce the likelihood of differential attrition by study arm. We created educational modules comparable in length and complexity of information to their corresponding intervention component but would not likely enhance psychological well-being outcomes (see Table 2). Inert content includes Cognitive Changes during Cancer, Nutrition, Weight Management, Financial Literacy, and Sun Protection.

Participants accessing the EMPOWER components or the Inert Content on the platform have a tab called "Discussion" where they can discuss topics related to the study. These topics are based on the Skills or Inert Content. For example, there is a discussion thread called "Gratitude" where participants can post their comments/thoughts on practicing gratitude. The discussion threads are moderated by a staff member on the research team to ensure the discussions stay on-topic, and to address or respond to signs of clinically significant distress, including suicide risk. All posts are tied to an EMPOWER Study ID and an avatar. No identifiable information is posted/shared. All participants have the opportunity to share only the content that they wish to share with the discussion boards. In addition, participants can

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opt in to share their home practice responses to the discussion board. All participants can reply, like, and thank individual posts. The discussion boards are only accessible to study participants and study team members.

Participants accessing the "Online Lessons" on the platform have a tab called "Awards" where they receive virtual badges for reaching certain goals (e.g., logging in daily for 5, 10, 15, 20, and 30 days, completing home practice for four consecutive days, or completing an online module). Each participant can see their earned badges as an incentive to proceed and engage with the study.

## 2.5. Study Measures

Participants complete a set of reliable and valid self-report questionnaires via REDCap at baseline (T0), weekly during the intervention (T1-T5), at intervention completion (T6), 2 months post intervention (T7), and 4 months post intervention (T8). The T0, T6, T7, and T8 assessments each last approximately 20 minutes and participants are compensated $25 for each assessment. During the intervention, participants complete a brief questionnaire at the end of each week (T1-T5) and are compensated $5 for each assessment. The brief questionnaires take approximately five minutes to complete. All measures are completed remotely via participants' personal computers, tablet, or smartphones. The maximum compensation for the study is $125. In addition to the measures listed in Table 3, key demographics (race/ethnicity, education, household income, insurance status) and clinical characteristics (cancer type, time since diagnosis, type of treatment, and time since treatment) are self-reported at baseline.

**2.5.1. Primary Outcome—**The primary outcome for this study is positive affect, which reflects the experiential aspect of psychological well-being. Positive affect is assessed at all timepoints throughout the intervention (T0-T8) and measured by the PROMIS Positive Affect computer adaptive test (CAT) [56]. Sample items include "I felt happy", "I felt delighted", "I felt peaceful." For PROMIS measures, a score of 50 is the average for the U.S. general population with a standard deviation of 10. Higher scores indicate more of the underlying construct. In this case, higher scores suggest a greater experience of positive emotions.

**2.5.2. Secondary Outcomes—**The secondary outcomes are measures of psychological well-being that capture both the negative experiential (anxiety, depression), positive evaluative (life satisfaction, meaning & purpose), and global aspects of emotional health. These are measured using the following scales: PROMIS Anxiety and Depression CATs [57], PROMIS Life Satisfaction CAT [56], PROMIS Meaning & Purpose CAT [58], and PROMIS Global Health (Mental) [59]. Anxiety and depression are assessed at all timepoints (T0-T8). Life satisfaction, meaning & purpose, and global mental health are assessed at baseline (T0), post-intervention (T6), 2-month follow-up (T7) and 4-month follow-up (T8).

**2.5.3. Exploratory Outcomes—**The exploratory outcomes include adherence, retention, physical functioning, fatigue, satisfaction with social roles and responsibilities, global physical health, and health behaviors. Adherence is calculated based on the number

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of intervention or inert sessions completed, the frequency of completing skills practices, and number of website visits. Retention is based on the proportion of participants who provide T6, T7, and T8 data. Additionally, we are exploring engagement based on the number of discussion board posts and frequency and types of badges received. AYAs who participate in the discussion board are asked to complete a series of questions to rate the relative usefulness of this EMPOWER feature, the most to least helpful features of the discussion board, and the relative importance of potential new features in future studies.

PROMIS measures are used for assessing AYAs' physical functioning [60], fatigue [61], satisfaction with social roles and responsibilities [62], and global physical health [59]. Healthy behaviors often associated with enhanced coping and better adjustment [63] are assessed. We use items from the Health Information National Trends Survey (HINTS) to assess diet, alcohol consumption, and cigarette smoking [64]. We assess exercise with the Godin Leisure Time Exercise Questionnaire [65] and sedentary behavior with the Marshall Sitting Questionnaire [66]. All of the exploratory, self-report measures are assessed at baseline (T0), post-intervention (T6), 2-month follow-up (T7), and 4-month follow-up (T8).

**2.5.4. Mediating and Moderating Variables—**A potential mediating variable identified by Fredrickson's Broaden and Build Theory [48] is coping self-efficacy. Potential moderating variables are social support, general self-efficacy, weekly stressors, and weekly positive events. We administer the PROMIS Self-Efficacy for Managing Emotions CAT [67], the PROMIS Emotional Support CAT [68], the PROMIS General Self-Efficacy CAT [69], self-efficacy for health behavior questions (e.g., "I can maintain a healthy weight", "I can maintain a healthy diet"), the Inventory of Stressful Events [70], and the number of positive events [70] to assess these respective constructs. Coping self-efficacy and general self-efficacy are assessed at all timepoints (T0-T8). Social support is assessed at baseline (T0), post-intervention (T6), 2-month follow-up (T7) and 4-month follow-up (T8). Weekly stressors and weekly positive events are assessed during the five weeks of the intervention (T1-T5), and self-efficacy for health behavior items are assessed pre- (T0) and post-intervention (T6).

## 3. Statistical Considerations

### 3.1. Analysis of Primary Objective

The primary outcome is positive affect at the conclusion of each intervention component. We will use a mixed model adjusted for baseline to model the value of the primary outcome (positive affect) at each time point following the intervention (T1-T8), including an indicator variable for each weekly intervention component received at T1 to T5, each time, and the two-way interaction of time with the corresponding indicator variable for the immediate post intervention time (e.g., T6). We will also evaluate interactions among the intervention components at T1 to T5. This model should allow us to isolate when the changes in the outcomes occur, and to examine if effects are stronger immediately after the content is delivered. We included the T7 and T8 time points to provide more precise variance estimation, but the post-intervention time points (T1-T6) will be the primary time points of interest. The main effects of each intervention component will be estimated from

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this model using contrasts. Due to randomization, we do not anticipate any differences in covariates between groups; however, we will evaluate differences by main effects and adjust if necessary. Effect sizes will be estimated using baseline standard deviations. Our pilot research suggests 0.30 is the minimally important difference among AYAs [45] and so this will be used as our criterion for retaining individual intervention components in the final optimized EMPOWER intervention.

### 3.2. Analysis of Secondary Objectives

We will use the same approach used for the primary outcome to model the effect of the intervention components on each of the secondary outcomes of depression and anxiety. For the analysis of the impact of the intervention on the secondary outcomes of life satisfaction, meaning & purpose, and global mental health, we will use the T0, T6, T7, and T8 measures. An ANCOVA model with indicator variables for each intervention component received at T1 to T5 will be used to model the impact of the intervention, including the baseline model and the time effect in the model.

### 3.3. Analysis of Exploratory Objectives

To summarize adherence and retention, we will calculate adherence as the number of intervention or inert sessions completed (maximum of 5), the frequency of completing skills practices (maximum of 35), and number of website visits (no *a priori* maximum). We will calculate and report the mean adherence across all individuals as well as the proportion of participants who completed three or more sessions. In previous studies of the online, self-guided intervention, these multiple operationalizations of adherence were highly correlated with each other [71]. Retention is defined as the proportion of participants who provide T6, T7, and T8 data. Participants who discontinue the intervention but complete the outcome assessments are counted in the numerator for calculating retention. Retention estimates will be calculated overall and by site.

We will use the same model described for the secondary outcome of life satisfaction to the exploratory outcomes of physical functioning, fatigue, satisfaction with social roles and responsibilities, diet, exercise, sedentary behavior, alcohol consumption, and cigarette smoking.

Coping self-efficacy is measured at T0-T8. To examine its role as a mediator, first we will examine how it changes over time and by intervention group, using the modeling approach described for the primary outcome. Next, we will fit a mixed model to model the impact of coping self-efficacy, time, and their interaction on the outcome of interest. Finally, we will include it in the models for the primary and secondary outcomes (with the intervention group indicators) including it as a time varying covariate.

Moderation analyses will be conducted by including the interaction of the potential moderating variables (age, gender, social support, stressors, and positive events) with the intervention group indicators. Moderation will be used to help identify those AYAs who are most “at risk” of poorer outcomes and could thus benefit most from the intervention. For age, gender, and social support, the baseline value will be used. Stressors and positive events will be treated as time varying covariates.

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### 3.4. Power and Sample Size

The MOST framework is a resource-efficient strategy that utilizes all participants to calculate main effects. Unlike a classical multi-arm RCT, the full-factorial experiment preserves power, which is driven by the total sample size, not cell size.[52] Sample size estimates are based on the lowest meaningful effect size for intervention components, ensuring the sample size is adequate to detect larger effect sizes.[53] Based on the MOST framework presented in Figure 2, individual intervention components that demonstrate an effect during Step 3 (e.g., ≥0.30) are included in intervention assembly and confirmatory testing in Steps 4–5. Therefore, we have set alpha at 0.10 given our tolerance for Type I error because intervention components will go on for confirmatory testing in future trials. Power analysis for the full factorial experiment is based on a total of n=352 recruited participants and a 72% retention rate at 1-month follow-up (final n=256). The maximum detectable effect size for a main effect is 0.31 (95% CI= (0.06, 0.56)) at 80% power with no correlation between the pretest and posttest (which is highly unlikely). For this scenario, the detectable effect size for an interaction is 0.62 (95% CI= (0.37, 0.87)). As the correlation between pretest and posttest increases, the effect size that we can detect decreases. This implies that the study has sufficient power to pinpoint an effective intervention component that is effective for inclusion into the optimized design. The study may not have sufficient power to detect small interaction (synergistic) effects (<0.37–0.62). The power analysis was conducted using the SAS macro FactorialPowerPlan.[72]

## 4. Discussion and Conclusion

The lack of evidence-based, scalable psychosocial interventions to improve psychological wellbeing, especially indicators of *positive* psychological wellbeing of AYA cancer survivors, limits the field's ability to address the mental health and quality of life needs of this large, underserved population. Building from our pilot study of EMPOWER, a multi-component, web-based, psychological well-being intervention for AYAs [45], the main goal of this project is to collect new data within an efficient research design to identify which intervention components are most beneficial and for which subgroups of AYAs are most "at risk" in order to inform identification and allocation of targeted supportive care resources. These data will guide future testing of the components of EMPOWER with the greatest likelihood of positive outcomes for AYAs treated in community-based settings, where access to mental health care remains limited, despite high need.

Our proposed study has significant strengths. First, psychosocial interventions that specifically target positive aspects of psychological well-being are infrequently tested in cancer survivorship despite their potential beneficial effects. In our meta-analyses of interventions that impact positive affect outcomes in cancer, the average effect of psychosocial interventions was modest (g=0.35) [73] but comparable to effects of interventions to manage common negative psychosocial sequelae in cancer such as pain (g=0.34 for pain severity, 0.40 for pain interference) [74], fatigue (g=0.26 to 0.30 across various interventions) [75], and depression (g=0.43) [76] and effects of positive psychological interventions more broadly (e.g., g=0.34) [77]. Our dual approach that focuses on decreasing the "negative" and enhancing the "positive" aspects of well-being will allow

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us to impact mental health by reducing and shortening psychological distress as well as increasing and sustaining psychological well-being.

Additional strengths include the mode of intervention delivery, innovative research design, and choice of patient-reported outcome measures. EMPOWER is a web-based intervention that is platform neutral, allowing it to be accessible via smartphone (both iPhone and Android systems), tablet PC, or desktop PC. Leveraging AYAs' technological aptitude for multi-component, tailored intervention delivery allows us to match their needs and preferences to supportive care content. Moreover, since EMPOWER is scalable, it can be simultaneously delivered to a limitless number of AYAs at multiple and geographically diverse sites. Treatment integrity and fidelity to EMPOWER remain fully intact, reducing threats to internal validity. Thus, there is great long-term potential to reach AYAs who are underserved and might not typically have access to psychosocial services through community-based practices where a majority receive care[78, 79]. In addition, the MOST framework is innovative and particularly useful for building, optimizing, and evaluating multicomponent behavioral interventions [52]. Having already built our individual intervention components [45], we will evaluate and optimize our EMPOWER intervention by assembling only those components that are expected to be efficacious. The optimized EMPOWER intervention will then be ready for confirmatory testing through a future RCT. Lastly, our proposal utilizes the state of the art in patient-reported outcomes by including emotional, physical, and social health measures from the NIH PROMIS [56–58, 61, 69, 80]. These psychometrically robust measures have been systematically created through rigorous qualitative and quantitative methodologies, yielding reliable, valid, and responsive patient-reported outcomes. Moreover, the ability to administer these measures as CATs minimizes respondent burden without sacrificing measurement precision. Thus, we can assess more content-relevant domains with fewer questions.

Upon completion of this study, we will have data to inform a fully optimized intervention (EMPOWER) to improve psychological well-being among AYA cancer survivors by reducing distress (depression and anxiety) and enhancing well-being (positive affect and life satisfaction). Moreover, we will be well-positioned to submit a larger, multi-site RCT by leveraging the NCI Community Oncology Research Program (NCORP). A subsequent trial would serve as the confirmatory RCT of our web-based, scalable EMPOWER intervention for AYA cancer survivors. Future work will also benefit from examining best practices for dissemination and implementation of EMPOWER in order to care for AYAs who would otherwise not have access to supportive care interventions to help manage post-treatment distress and enhance well-being.

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## References

1. American Cancer Society, Cancer Facts & Figures 2020. 2020, American Cancer Society: Atlanta, GA.
2. Page LL, et al. , Prevalence of cancer survivors diagnosed during adolescence and young adulthood in the United States. J Natl Cancer Inst, 2024.
3. Barr RD, et al. , Cancer in Adolescents and Young Adults: A Narrative Review of the Current Status and a View of the Future. JAMA Pediatr, 2016. 170(5): p. 495–501. [PubMed: 26999630]
4. Tai E, et al. , Health status of adolescent and young adult cancer survivors. Cancer, 2012. 118(19): p. 4884–91. [PubMed: 22688896]
5. Kaul S, et al. , Cigarette smoking, comorbidity, and general health among survivors of adolescent and young adult cancer. Cancer, 2016. 122(18): p. 2895–905. [PubMed: 27286172]
6. Chao C, et al. , Cardiovascular Disease Risk Profiles in Survivors of Adolescent and Young Adult (AYA) Cancer: The Kaiser Permanente AYA Cancer Survivors Study. J Clin Oncol, 2016. 34(14): p. 1626–33. [PubMed: 26951318]
7. Chao C, et al. , Chronic Comorbidities Among Survivors of Adolescent and Young Adult Cancer. J Clin Oncol, 2020. 38(27): p. 3161–3174. [PubMed: 32673152]
8. Bank TW Life expectancy at birth, total (years). November 1, 2018]; Available from: http://data.worldbank.org/indicator/SP.DYN.LE00.IN.
9. Zebrack B, Information and service needs for young adult cancer survivors. Supportive Care in Cancer, 2009. 17(4): p. 349–57. [PubMed: 18543006]
10. Zebrack BJ, Psychological, social, and behavioral issues for young adults with cancer. Cancer, 2011. 117(10): p. 2289–94. [PubMed: 21523748]
11. Smith AW, et al. , Health-related Quality of Life of Adolescent and Young Adult Cancer Patients in the United States: the AYA HOPE study. Journal of Clinical Oncology, 2013. 31(17): p. 2136–45. [PubMed: 23650427]
12. Husson O, et al. , Adolescent and young adult (AYA) lymphoma survivors report lower health-related quality of life compared to a normative population: results from the PROFILES registry. Acta Oncol, 2017. 56(2): p. 288–294. [PubMed: 28077017]
13. Salsman JM, et al. , Physical, emotional, and social health differences between post-treatment young adults with cancer and matched healthy controls. Cancer, 2014. 120(15): p. 2247–2254. [PubMed: 24888335]
14. Schulte FSM, et al. , Quality of life among survivors of adolescent and young adult cancer in Canada: A Young Adults With Cancer in Their Prime (YACPRIME) study. Cancer, 2021. 127(8): p. 1325–1333. [PubMed: 33332603]
15. Rotz SJ, et al. , Health-Related Quality of Life in Young Adult Survivors of Hematopoietic Cell Transplantation. Transplant Cell Ther, 2022. 28(10): p. 701.e1–701.e7.
16. Park EM and Rosenstein DL, Depression in adolescents and young adults with cancer. Dialogues in clinical neuroscience, 2015. 17(2): p. 171. [PubMed: 26246791]
17. Sansom-Daly UM and Wakefield CE, Distress and adjustment among adolescents and young adults with cancer: an empirical and conceptual review. Transl Pediatr, 2013. 2(4): p. 167–97. [PubMed: 26835313]
18. Lang MJ, David V, and Giese-Davis J, The Age Conundrum: A Scoping Review of Younger Age or Adolescent and Young Adult as a Risk Factor for Clinical Distress, Depression, or Anxiety in Cancer. J Adolesc Young Adult Oncol, 2015. 4(4): p. 157–73. [PubMed: 26697266]
19. Smith AW, et al. , Next steps for adolescent and young adult oncology workshop: An update on progress and recommendations for the future. Cancer, 2016. 122(7): p. 988–999. [PubMed: 26849003]

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20. Salsman JM and Rosenberg AR, Fostering resilience in adolescence and young adulthood: Considerations for evidence-based, patient-centered oncology care. Cancer, 2024. 130(7): p. 1031–1040. [PubMed: 38163249]
21. Murphy KM, et al. , A systematic review of health-related quality of life outcomes in psychosocial intervention trials for adolescent and young adult cancer survivors. Crit Rev Oncol Hematol, 2023. 188: p. 104045. [PubMed: 37269881]
22. Zhang A, et al. , Psychosocial, behavioral, and supportive interventions for pediatric, adolescent, and young adult cancer survivors: A systematic review and meta-analysis. Crit Rev Oncol Hematol, 2021. 160: p. 103291. [PubMed: 33675909]
23. McGrady ME, et al. , Psychological Outcomes in Adolescent and Young Adult Cancer Survivors. J Clin Oncol, 2024. 42(6): p. 707–716. [PubMed: 37967297]
24. Park CL, et al., Emotional Well-Being: What It Is and Why It Matters. Affective Science, 2022.
25. Pressman SD and Cohen S, Does positive affect influence health? Psychological Bulletin, 2005. 131(6): p. 925–971. [PubMed: 16351329]
26. Chida Y and Steptoe A, Positive Psychological Well-Being and Mortality: A Quantitative Review of Prospective Observational Studies. Psychosomatic Medicine, 2008. 70(7): p. 741–756. [PubMed: 18725425]
27. Moskowitz JT, Positive Affect Predicts Lower Risk of AIDS Mortality. Psychosomatic medicine, 2003. 65(4): p. 620. [PubMed: 12883113]
28. Steptoe A and Wardle J, Positive affect measured using ecological momentary assessment and survival in older men and women. Proceedings of the National Academy of Sciences, 2011. 108(45): p. 18244–18248.
29. Moskowitz JT, Acree M, and Epel ES, Positive Affect Uniquely Predicts Lower Risk of Mortality in People With Diabetes. Health Psychology, 2008. 27(1 Suppl.): p. S73–S82. [PubMed: 18248108]
30. Liu B, et al. , Does happiness itself directly affect mortality? The prospective UK Million Women Study. Lancet, 2016. 387(10021): p. 874–81. [PubMed: 26684609]
31. Lyubomirsky S, King L, and Diener E, The benefits of frequent positive affect: does happiness lead to success? Psychol Bull, 2005. 131(6): p. 803–55. [PubMed: 16351326]
32. Rosenberg AR, et al. , Hope and benefit finding: Results from the PRISM randomized controlled trial. Pediatr Blood Cancer, 2019. 66(1): p. e27485. [PubMed: 30270489]
33. Steineck A, et al. , A Psychosocial Intervention's Impact on Quality of Life in AYAs with Cancer: A Post Hoc Analysis from the Promoting Resilience in Stress Management (PRISM) Randomized Controlled Trial. Children (Basel), 2019. 6(11).
34. Rosenberg AR, et al. , Promoting resilience in adolescents and young adults with cancer: Results from the PRISM randomized controlled trial. Cancer, 2018. 124(19): p. 3909–3917. [PubMed: 30230531]
35. Rosenberg AR, et al. , Assessment of the Promoting Resilience in Stress Management Intervention for Adolescent and Young Adult Survivors of Cancer at 2 Years: Secondary Analysis of a Randomized Clinical Trial. JAMA Netw Open, 2021. 4(11): p. e2136039. [PubMed: 34817581]
36. Chandeying N and Thongseiratch T, Online Interventions to Improve Mental Health of Pediatric, Adolescent, and Young Adult Cancer Survivors: A Systematic Review and Meta-Analysis. Front Psychiatry, 2021. 12: p. 784615. [PubMed: 35002804]
37. Zhang A, et al. , Technology-Assisted Psychosocial Interventions for Childhood, Adolescent, and Young Adult Cancer Survivors: A Systematic Review and Meta-Analysis. J Adolesc Young Adult Oncol, 2022. 11(1): p. 6–16. [PubMed: 33960845]
38. Lim J. w., Zhang A, and Zebrack B, Technology-Based Cancer Survivorship Care Interventions: A Systematic Review and Meta-Analysis. Research on Social Work Practice, 2024. 34(7): p. 757–780.
39. Devine KA, et al. , Digital Health Interventions for Adolescent and Young Adult Cancer Survivors. JCO Clin Cancer Inform, 2018. 2: p. 1–15.
40. Rabin C, et al. , Intervention format and delivery preferences among young adult cancer survivors. Int J Behav Med, 2013. 20(2): p. 304–10. [PubMed: 22328444]

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

41. Painter JM, et al. , A positive emotion regulation intervention for bipolar I disorder: Treatment development and initial outcomes. Gen Hosp Psychiatry, 2019. 61: p. 96–103. [PubMed: 31439286]
42. Moskowitz JT, et al. , Randomized controlled trial of a positive affect intervention for people newly diagnosed with HIV. J Consult Clin Psychol, 2017. 85(5): p. 409–423. [PubMed: 28333512]
43. Moskowitz JT, et al. , Randomized controlled trial of a facilitated online positive emotion regulation intervention for dementia caregivers. Health Psychol, 2019. 38(5): p. 391–402. [PubMed: 31045422]
44. Cheung EO, et al. , A randomized pilot trial of a positive affect skill intervention (lessons in linking affect and coping) for women with metastatic breast cancer. Psychooncology, 2017. 26(12): p. 2101–2108. [PubMed: 27862646]
45. Salsman JM, et al. , An eHealth, Positive Emotion Skills Intervention for Enhancing Psychological Well-Being in Young Adult Cancer Survivors: Results from a Multi-Site, Pilot Feasibility Trial. Int J Behav Med, 2023.
46. Carr A, et al. , Effectiveness of positive psychology interventions: a systematic review and meta-analysis. The Journal of Positive Psychology, 2021. 16(6): p. 749–769.
47. Lyubomirsky S and Layous K, How do simple positive activities increase well-being? Current Directions in Psychological Science, 2013. 22(1): p. 57–62.
48. Fredrickson BL, The Role of Positive Emotions in Positive Psychology: The Broaden- and-Build Theory of Positive Emotions. American Psychologist, 2001. 56(3): p. 218–26. [PubMed: 11315248]
49. Folkman S, The case for positive emotions in the stress process. Anxiety Stress Coping, 2008. 21(1): p. 3–14. [PubMed: 18027121]
50. Moskowitz JT, Addington EL, and Cheung EO, Positive psychology and health: Well-being interventions in the context of illness. Gen Hosp Psychiatry, 2019. 61: p. 136–138. [PubMed: 31757566]
51. Collins LM, et al. , A strategy for optimizing and evaluating behavioral interventions. Ann Behav Med, 2005. 30(1): p. 65–73. [PubMed: 16097907]
52. Collins LM, Murphy SA, and Strecher V, The multiphase optimization strategy (MOST) and the sequential multiple assignment randomized trial (SMART): new methods for more potent eHealth interventions. Am J Prev Med, 2007. 32(5 Suppl): p. S112–8. [PubMed: 17466815]
53. Wyrick DL, et al. , Moving beyond the treatment package approach to developing behavioral interventions: addressing questions that arose during an application of the Multiphase Optimization Strategy (MOST). Transl Behav Med, 2014. 4(3): p. 252–9. [PubMed: 25264465]
54. REDCap Consortium. REDCap (Research Electronic Data Capture) Consortium. 2012 [cited 2012 January 17]; Available from: http://project-redcap.org/.
55. Harris PA, et al. , Research electronic data capture (REDCap)-A metadata-driven methodology and workflow process for providing translational research informatics support. Journal of Biomedical Informatics, 2009. 42(2): p. 377–381. [PubMed: 18929686]
56. Salsman J, et al. , Assessing psychological well-being: self-report instruments for the NIH Toolbox. Quality of Life Research, 2014. 23(1): p. 205–15. [PubMed: 23771709]
57. Pilkonis PA, et al. , Item banks for measuring emotional distress from the Patient-Reported Outcomes Measurement Information System (PROMIS):depression, anxiety, and anger. Assessment, 2011. 18(3): p. 263–83. [PubMed: 21697139]
58. Salsman JM, et al. , Assessing meaning & purpose in life: development and validation of an item bank and short forms for the NIH PROMIS(®). Qual Life Res, 2020. 29(8): p. 2299–2310. [PubMed: 32306302]
59. Hays RD, et al. , Development of physical and mental health summary scores from the Patient Reported Outcomes Measurement Information System (PROMIS) global items. Quality of Life Research, 2009. 18(7): p. 873–80. [PubMed: 19543809]
60. Fries JF, et al. , Progress in assessing physical function in arthritis: PROMIS short forms and computerized adaptive testing. Journal of Rheumatology, 2009. 36(9): p. 2061–6. [PubMed: 19738214]

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61. Garcia SF, et al. , Standardizing patient-reported outcomes assessment in cancer clinical trials: a patient-reported outcomes measurement information system initiative. Journal of Clinical Oncology, 2007. 25(32): p. 5106–12. [PubMed: 17991929]
62. Cook KF, et al. , Six Patient-Reported Outcome Measurement Information System Short Form Measures Have Negligible Age- or Diagnosis-Related Differential Item Functioning in Individuals With Disabilities. Archives of Physical Medicine and Rehabilitation, 2012. Epub ahead of print.
63. Doyle C, et al. , Nutrition and Physical Activity During and After Cancer Treatment: An American Cancer Society Guide for Informed Choices. CA, 2006. 56(6): p. 323. [PubMed: 17135691]
64. Nelson DE, et al. , The Health Information National Trends Survey (HINTS): development, design, and dissemination. J Health Commun, 2004. 9(5): p. 443–60; discussion 81–4. [PubMed: 15513791]
65. Godin G and Shephard RJ, A simple method to assess exercise behavior in the community. Canadian Journal of Applied Sport Sciences, 1985. 10(3): p. 141–6.
66. Marshall AL, et al. , Measuring total and domain-specific sitting: a study of reliability and validity. Med Sci Sports Exerc, 2010. 42(6): p. 1094–102. [PubMed: 19997030]
67. Gruber-Baldini AL, et al. , Validation of the PROMIS® measures of self-efficacy for managing chronic conditions. Quality of Life Research, 2017: p. 1–10.
68. HealthMeasures. HealthMeasures: Transforming how health is measured. 2017 [cited 2017 January 18]; Available from: http://www.healthmeasures.net/index.php.
69. Salsman JM, et al. , Calibration and initial validation of a general self-efficacy item bank and short form for the NIH PROMIS((R)). Qual Life Res, 2019. 28(9): p. 2513–2523. [PubMed: 31140041]
70. Almeida DM, Wethington E, and Kessler RC, The daily inventory of stressful events: an interview-based approach for measuring daily stressors. Assessment, 2002. 9(1): p. 41–55. [PubMed: 11911234]
71. Moskowitz JT, et al. , Facilitator Contact, Discussion Boards, and Virtual Badges as Adherence Enhancements to a Web-Based, Self-guided, Positive Psychological Intervention for Depression: Randomized Controlled Trial. J Med Internet Res, 2021. 23(9): p. e25922. [PubMed: 34550076]
72. Dziak J, Collins L, and Wagner A, FactorialPowerPlan SAS macro suite users' guide. 2013, University Park: The Methodology Center, Penn State.
73. Salsman JM, et al. , Psychosocial interventions for cancer survivors: A meta-analysis of effects on positive affect. J Cancer Surviv, 2019.
74. Sheinfeld Gorin S, et al. , Meta-analysis of psychosocial interventions to reduce pain in patients with cancer. J Clin Oncol, 2012. 30(5): p. 539–47. [PubMed: 22253460]
75. Mustian KM, et al. , Comparison of Pharmaceutical, Psychological, and Exercise Treatments for Cancer-Related Fatigue: A Meta-analysis. JAMA Oncol, 2017. 3(7): p. 961–968. [PubMed: 28253393]
76. Hart SL, et al. , Meta-analysis of efficacy of interventions for elevated depressive symptoms in adults diagnosed with cancer. J Natl Cancer Inst, 2012. 104(13): p. 990–1004. [PubMed: 22767203]
77. Weiss LA, Westerhof GJ, and Bohlmeijer ET, Can We Increase Psychological Well-Being? The Effects of Interventions on Psychological Well-Being: A Meta-Analysis of Randomized Controlled Trials. PLoS One, 2016. 11(6): p. e0158092. [PubMed: 27328124]
78. U.S. Department of Health and Human Services, et al. Closing the gap: research and care imperatives for adolescents and young adults with cancer. 2006 [cited 2008 April 6]; Available from: http://planning.cancer.gov/library/AYAO_PRG_Report_2006_FINAL.pdf.
79. Zebrack B, Mathews-Bradshaw B, and Siegel S, Quality cancer care for adolescents and young adults: a position statement. J Clin Oncol, 2010. 28(32): p. 4862–7. [PubMed: 20855821]
80. Cella D, et al. , The Patient-Reported Outcomes Measurement Information System (PROMIS) developed and tested its first wave of adult self-reported health outcome item banks: 2005–2008. Journal of Clinical Epidemiology, 2010. 63(11): p. 1179–1194. [PubMed: 20685078]

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**Highlights:**

- The MOST design will identify intervention components most likely to be beneficial.
- Our framework incorporates a dual focus on reducing distress and enhancing well-being.
- This is the largest psychosocial intervention trial of post-treatment AYA survivors.

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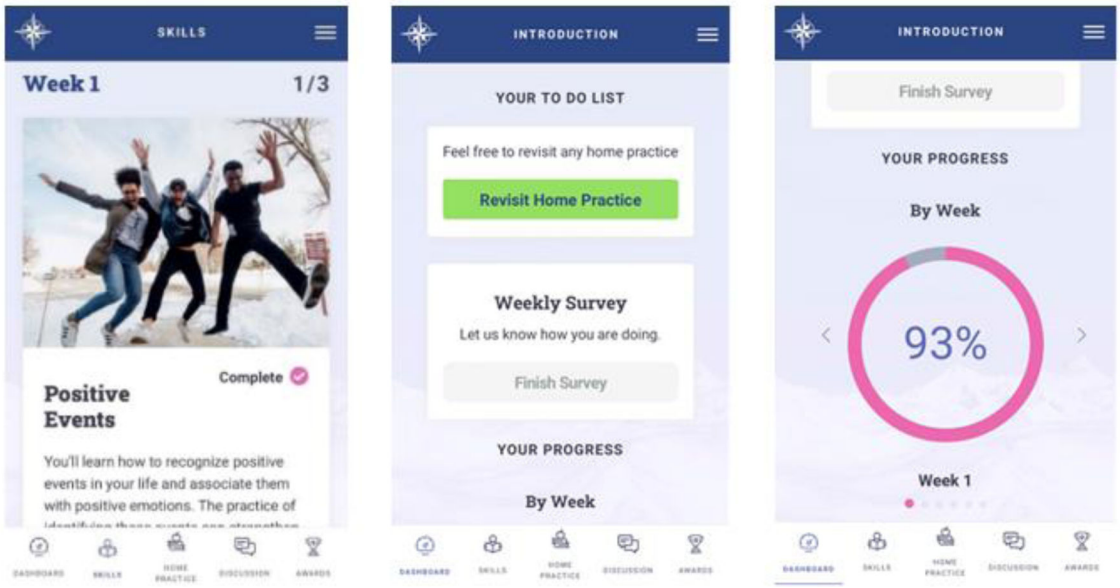


**Figure 1.**
Screenshots of the Mobile View of EMPOWER Content

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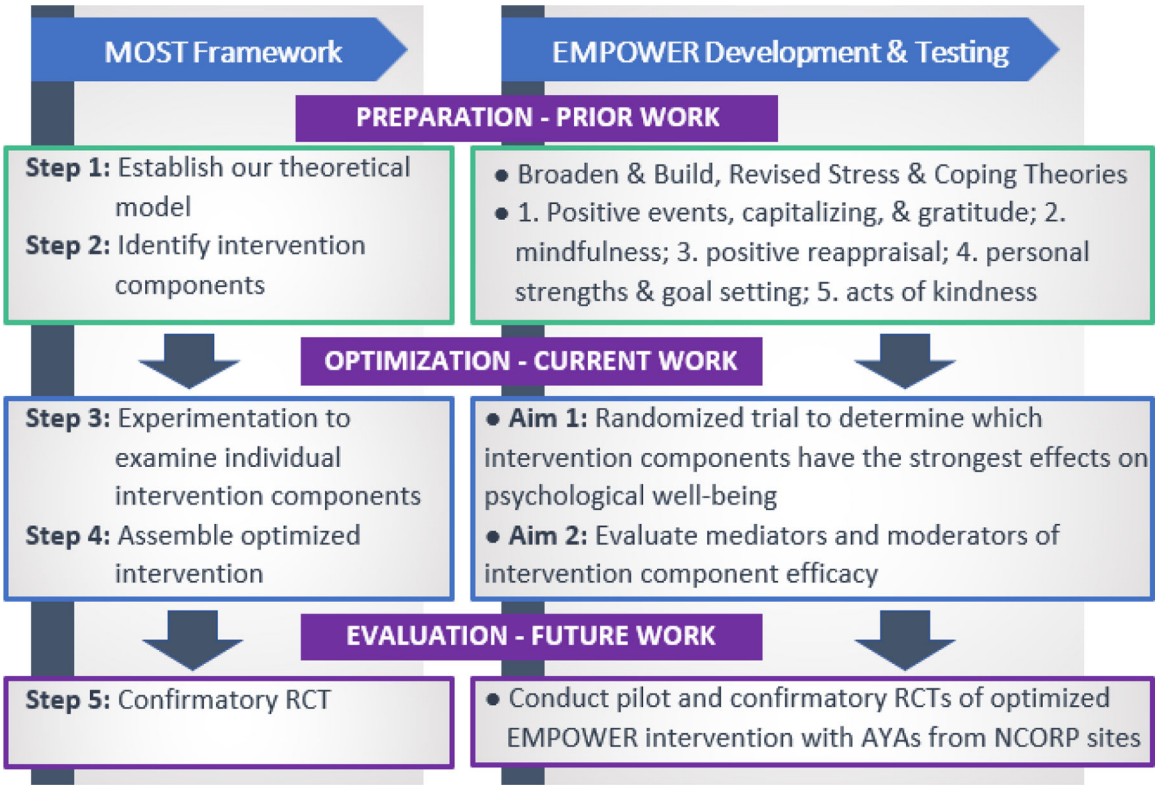


**Figure 2.**
The MOST Framework for EMPOWER Development & Testing

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**Table 1.**

Full Factorial Design for EMPOWER

| Conditions | Component 1 positive events capitalizing gratitude | Component 2 mindfulness | Component 3 positive reappraisal | Component 4 personal strengths goal setting | Component 5 acts of kindness |
|---|---|---|---|---|---|
| 1 | N | N | N | N | N |
| 2 | N | N | N | N | Y |
| 3 | N | N | N | Y | N |
| 4 | N | N | N | Y | Y |
| 5 | N | N | Y | N | N |
| 6 | N | N | Y | N | Y |
| 7 | N | N | Y | Y | N |
| 8 | N | N | Y | Y | Y |
| 9 | N | Y | N | N | N |
| 10 | N | Y | N | N | Y |
| 11 | N | Y | N | Y | N |
| 12 | N | Y | N | Y | Y |
| 13 | N | Y | Y | N | N |
| 14 | N | Y | Y | N | Y |
| 15 | N | Y | Y | Y | N |
| 16 | N | Y | Y | Y | Y |
| 17 | Y | N | N | N | N |
| 18 | Y | N | N | N | Y |
| 19 | Y | N | N | Y | N |
| 20 | Y | N | N | Y | Y |
| 21 | Y | N | Y | N | N |
| 22 | Y | N | Y | N | Y |
| 23 | Y | N | Y | Y | N |
| 24 | Y | N | Y | Y | Y |
| 25 | Y | Y | N | N | N |
| 26 | Y | Y | N | N | Y |
| 27 | Y | Y | N | Y | N |
| 28 | Y | Y | N | Y | Y |
| 29 | Y | Y | Y | N | N |
| 30 | Y | Y | Y | N | Y |
| 31 | Y | Y | Y | Y | N |
| 32 | Y | Y | Y | Y | Y |

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**Table 2.**

Weekly EMPOWER and Inert Content

| | EMPOWER | | Inert Content | |
|---|---|---|---|---|
| **Session** | **Skills** | **Examples** | **Skills** | **Examples** |
| Week 1 | Positive Events Capitalizing Gratitude | Learning to recognize positive events Amplifying the experience of positive events Feeling thankful for your life | Cognitive Changes during Cancer | Practicing strategies to improve attention and concentration |
| Week 2 | Mindfulness | Practicing awareness and non-judgmental | Nutrition | Maintaining a regular food diary |
| Week 3 | Positive Reappraisal | Understanding different forms of positive reappraisal in the midst of stress | Weight Management | Calculating person Body Mass Index and base metabolic rate |
| Week 4 | Personal Strengths Achievable Goals | Listing and noting use of personal strengths Understanding and setting attainable goals | Financial Literacy | Learning basic elements of budgeting |
| Week 5 | Acts of Kindness | Understanding the impact of small acts of kindness | Sun Protection | Practicing using sunscreen effectively |

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**Table 3.**

EMPOWER Study Measures

| Variable(s) | Measure(s) | Timepoint Administered | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | T0 | T1 | T2 | T3 | T4 | T5 | T6 | T7 | T8 |
| *Primary Outcome* | | | | | | | | | | |
| Psychological Well-Being | PROMIS Positive Affect CAT | X | X | X | X | X | X | X | X | X |
| *Secondary Outcomes* | | | | | | | | | | |
| | PROMIS Anxiety CAT | X | X | X | X | X | X | X | X | X |
| | PROMIS Depression CAT | X | X | X | X | X | X | X | X | X |
| | PROMIS Life Satisfaction CAT | X | | | | | | X | X | X |
| | PROMIS Meaning &Purpose CAT | X | | | | | | X | X | X |
| | NIH PROMIS Global Health (Mental) | X | | | | | | X | X | X |
| *Exploratory Outcomes* | | | | | | | | | | |
| Physical Well-Being | PROMIS Physical Functioning CAT | X | | | | | | X | X | X |
| | PROMIS Fatigue CAT | X | | | | | | X | X | X |
| | PROMIS Global Health (Physical) | X | | | | | | X | X | X |
| Social Well Being | PROMIS Satisfaction with Social Roles CAT | X | | | | | | X | X | X |
| Health Behaviors | Diet | X | | | | | | X | X | X |
| | Alcohol Use & Smoking | X | | | | | | X | X | X |
| | Godin Leisure Time Exercise Questionnaire | X | | | | | | X | X | X |
| | Marshall Sitting Questionnaire | X | | | | | | X | X | X |
| *Mediating Variable* | | | | | | | | | | |
| Self-Efficacy | PROMIS Self-Efficacy for Managing Emotions CAT | X | X | X | X | X | X | X | X | X |
| | Self-Efficacy Questions | X | | | | | | X | | |
| *Moderating Variable* | | | | | | | | | | |
| Social Support | PROMIS Emotional Support CAT | X | | | | | | X | X | X |
| General Self-Efficacy | PROMIS General Self-Efficacy CAT | X | X | X | X | X | X | X | X | X |
| *Other Measures* | | | | | | | | | | |
| Demographics | Age, gender, race, ethnicity, education, income, job status, relationship status, and insurance | X | | | | | | | | |
| Clinical | Cancer diagnosis, stage, treatment type, performance status | X | | | | | | | | |

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| Variable(s) | Measure(s) | Timepoint Administered | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | T0 | T1 | T2 | T3 | T4 | T5 | T6 | T7 | T8 |
| Weekly Stressors & Positive Events | Inventory of Stressful Events | | X | X | X | X | X | | | |

Note: T0 = baseline/pre-intervention; T1 to T5 = Week 1 to Week 5 during the intervention, each assessment occurs after the assigned, weekly component; T6=post-intervention, T7=2 month follow-up; T8=4-month follow-up.